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An Evolutionary Perspective on Dog Behavioral Genetics

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dog, canine, behavior, ethology, evolution, genomics, genetics

Abstract

Dogs have played an outsized role in the field of behavioral genetics since its earliest days. Their unique evolutionary history and ubiquity in the modern world make them a potentially powerful model system for discovering how genetic changes lead to changes in behavior. Genomic technology has supercharged this potential by enabling scientists to sequence the DNA of thousands of dogs and test for correlations with behavioral traits. However, fractures in the early history of animal behavior between biological and psychological subfields may be impeding progress. In addition, canine behavioral genetics has included almost exclusively dogs from modern breeds, who represent just a small fraction of all dog diversity. By expanding the scope of dog behavior studies, and incorporating an evolutionary perspective on canine behavioral genetics, we can move beyond associations to understanding the complex interactions between genes and environment that lead to dog behavior.

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1. INTRODUCTION

Dogs, and dog behavior, have had an outsized influence on the field of behavioral genetics. Dogs exhibit a remarkable range of heritable and sometimes useful behaviors, and most people can observe this directly through regular interactions starting in childhood. People—including scientists—are fascinated by dogs. When Charles Darwin needed to explain how selection can change populations in *On the Origin of Species*, he used dogs as an example (1). However, our close relationship with dogs can also limit our ability to act as impartial observers and impede research into how genetics shapes canine behavior. Further complicating this is an historic division between ethology, which focuses on the evolution of animal behavior, and comparative psychology, which focuses on how animals learn during their lifetime. The advent of genomics offers an opportunity to investigate questions in canine behavioral genetics using extremely powerful new technologies. It is essential that we avoid overly simplistic interpretations of dog behavior and start from the premise that behavior in dogs, like in every other species, is a complex interaction of genes and environment. We highlight how integrating expertise in animal behavior will accelerate the growth of canine behavioral genetics and ultimately help us understand how genetics shapes behavior in other species, including ourselves.

2. THE HISTORY OF DOGS

Dogs are wolves who rapidly evolved to survive and thrive in the human niche. As a result of this evolutionary process, dogs differ from wolves in many ways, including in their behavior (2). Dogs have lived in and around human settlements for well over 12,000 years (3–5). To survive as scavengers of human food and waste, they evolved foraging, hazard avoidance, and reproductive behaviors useful for their niche as scavengers of human waste (6–10) and lost hunting behaviors that in wolves are essential for survival (2). Domestication-related behaviors are those that are common to all dogs and differentiate them from wolves. At some point, dogs started being used for tasks like guarding livestock and, later on, herding livestock and hunting game. These working behaviors are repurposed versions of the wolf predatory sequence (11). They vary between dogs and are highly heritable (**Figure 1**).

Early humans may have inadvertently selected on behavior by giving favored dogs increased access to limited resources, but there is little evidence of humans intentionally breeding dogs until approximately 2,000 years ago (12). Modern dog breeds are less than 160 years old (~50–80 generations). They are an invention of Victorian England, where dog fanciers began breeding animals with desired traits together to achieve a physical ideal and subsequently closed off the population to maintain the "purity" of the lineage (13, 14). Thus, all dogs of a given breed were descended directly from only those dogs in this final group or "stud book." Until that point, dogs had been defined as herders or retrievers based on the job they performed and not because of their parentage.

2.1. Avoiding Misconceptions About Canid Evolution in Study Design

The history of dogs is an important consideration in the design of behavioral genetic studies. Studies of domestication will sometimes assume that modern gray wolves are the wild ancestors of dogs, but in truth they are their closest living wild relative, which is an important distinction. They are descended from a more distant lineage of wolves than the dog progenitor population, which is unknown but likely extinct (15). Thus, differences between modern wolves and dogs are not necessarily related to domestication. The timeline of dog domestication also dispels a common misperception that some breeds are more ancient or more primitive [and therefore more "wolf-like" behaviorally and genetically (16, 17)]. All dogs are descended from the ancient domestic dogs, and no modern breed is more than a few centuries old. Dogs from breeds like the Finnish spitz

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Figure 1

Working behaviors in dogs are made up of pieces of the wolf predatory sequence. (*a*) Hunting dog showing orient, (*b*) terrier showing grab-bite, (*c*) herding dog showing upright herding body posture and eye (head at or below the level of the body), (*d*) herding dog showing eye and a creeping stalk (bent legs creeping forward), (*e*) sighthound showing chase, and (*f*) herding dog showing eye and frozen stalk (legs bent and holding still). Photos provided by (*a*,*b*,*e*) T. Sladden (copyright 2024 T. Sladden, CC BY 4.0), (*c*) A. Kendell (copyright 2024 A. Kendell, CC BY-NC-SA 4.0), (*d*) K. Mayfield (CC BY 4.0), and (*f*) A. Powell (CC BY 4.0).

and shar-pei are simply more genetically isolated, and thus less similar to other modern breeds, but not more similar to wolves (18).

2.2. Expanding the Diversity of Dog Research

To date, nearly all research on canine behavior has focused on modern dog breeds, even though they represent just a tiny fraction of the worldwide dog population. That includes most of the work reviewed here. However, and importantly, this excludes more than 80% of the nearly 1 billion dogs on Earth, who are free-living, free-breeding, and not under human control (e.g., village dogs) (2). It also limits the diversity of the study populations, and thus our ability to understand how these behaviors evolved and to discover the underlying molecular mechanisms. Even in breeds that were founded from free-living village dog populations (e.g., basenjis) or from landrace breeds (Great Pyrenees), most genetic diversity was lost in the population bottleneck that created the

modern breed from a handful of individuals (3). For example, modern Siberian huskies have far less genetic diversity than the ancestral sled dogs of Alaska (19). Huge populations of ancient American and South Pacific dogs with unique behavioral characteristics were supplanted by the less diverse European breed dogs, most likely during the colonization of these areas (20). Seeking out remnant populations of dogs with heritage from these largely extinct populations that survive in remote areas gives unique insights into dog genetics (21).

Although some breeds are enriched for dogs with a propensity toward behaviors like herding and retrieving, these ancient and complex behaviors long predate the breeds. Expanding the scope of canine research to consider all dogs (single, mixed, and non-breed dogs) would offer a more complete representation of dog behavioral and genetic diversity and facilitate new discoveries.

3. THE HISTORY OF DOG BEHAVIORAL GENETICS

3.1. Early History

From its inception in the first half of the 1900s, the modern field of animal behavior has come from two different disciplines: ethology and comparative psychology. Ethology emerged from the biological study of animal behavior and was concerned primarily with the evolution of animal behavior. Comparative psychology (as the name implies) originated from the larger field of psychology and was focused on understanding how animals learn behavior over their lifetimes, and how they compare with humans.

In the mid-1900s, ethology and comparative psychology became mired in what would later be called the nature versus nurture debate. Ethologists argued that nature (i.e., genetically controlled instincts) was the most important component of behavior, whereas comparative psychologists asserted that the key to animal behavior lay in nurture (i.e., environmentally controlled learning) (22). Eventually, all agreed that asking whether genes or environment was more important in determining behavior was a false dichotomy. They realized not only that both genes and environment were integral to behavior but that these two elements were not truly separable (23, 24).

Amidst this debate, John Paul Scott and John L. Fuller started the first and most well-known large-scale investigation of the relationship between genetics and behavior (25). Over 13 years, starting in 1945, they collected behavioral observations on hundreds of dogs produced through purposeful crossbreeding of five dog breeds that displayed significant variations in behavior (basenji, beagle, American cocker spaniel, Shetland sheepdog, and wire-haired fox terrier). Scott & Fuller's experimental setup at Jackson Laboratory allowed them to document behavior in a comprehensive, rigorous manner and at a scale that enabled them to be the first to generate quantitative estimates of heritability for behaviors in dogs.

Although Scott & Fuller's research aimed to investigate both genetic and environmental influences on dog behavior, their behavioral testing methods were adopted largely from the field of comparative psychology. As a result, Scott & Fuller's research was far more quantitative than prior studies, which relied on general observations of subjective behavioral concepts like temperament (such as fearfulness or willingness) or performance (such as judge scores in hunting field or Schutzhund trials). In contrast, Scott & Fuller measured behavior based on detailed observations and systematic, behavioral testing of individual dogs. For example, each pup underwent a battery of tests (emotional reactivity, barrier reactivity, training, breed-specific abilities, problem solving, and response to human handling) between the ages of 5 weeks and 35 weeks.

Scott & Fuller's (25) work, published in 1965, addressed the contribution of genetics to the production of mammalian behavior in unprecedented detail. They discovered that very young pups are not immature adults but rather highly adapted neonates, and they reframed development as the metamorphosis of a highly adapted neonate into a highly adapted adult. They addressed the

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still-ongoing nature versus nurture debate by exploring the effect of differential rearing on later social behavior and showing that inherited genetic factors interact with a pup's early environment to produce differences in adult behavior.

Although Scott & Fuller's (25) work was widely cited, the subtleties of their conclusions were often missed. For example, their work showed that within each dog breed, there was a large amount of individual variability in the behavior of dogs. Although there were some statistically significant differences in the means for each breed, the variability within the breed exceeded the variability between breeds.

Scott & Fuller's work reinforced the conclusion already reached by both comparative psychologists and ethologists that both genes and environment are key to animal behavior. Unfortunately, neither field retained this more holistic view in their research questions. Ethology and the fields of study that sprouted from it (e.g., behavioral ecology) continued to focus mainly on the heritable components of behavior and their evolution. Comparative psychology and the fields that sprung from it (e.g., cognitive psychology) continued to focus on the environmental components of behavior, and how behavior modification shapes the capability an animal can attain within its lifetime. Although some have attempted to bring nature and nurture together through subfields like developmental ecology (26), this work is the exception rather than the rule.

At the time of Scott & Fuller's work, it was not possible to connect genes to changes in behavior. DNA was not even confirmed to be genetic material until approximately seven years into their dog study (27). Despite many more subsequent studies of heritability of behavior (reviewed in 28), the field of behavioral genetics awaited a technological revolution.

3.2. The Advent of Dog Genomics and Gene Mapping

The 1950s marked the start of a pre-genomic era of rapid scientific discovery and technological development that culminated in the publication of the first human genome sequence in 2001 (29). A genome is the genetic information each individual inherits from its parents. Genomes are made of DNA, which is a molecule consisting of a very long chain of four different nucleotides: adenine, thiamine, cytosine, and guanine. The dog (and wolf) genome is approximately 2.4 billion nucleotides long, and each individual carries two copies, one from their father and one from their mother, for a total of 4.8 billion nucleotides. In most parts of their DNA sequence, dog's nucleotides will look identical (the parts that make them dogs), but small differences in DNA make each dog unique.

In dog breeds, all the dogs are very closely related to each other (much like a very large human family). Thus, when we compare the genomes of two dogs from the same breed, we find approximately tenfold fewer differences than when we compare two dogs from different breeds (14). The lower levels of genetic diversity are both a strength and a weakness in genetic studies.

Today, genome sequencing (reading all 4.8 billion nucleotides in a dog's genome) is routine, but sequencing the very first genomes cost billions of dollars. Still, the Dog Genome Project, launched in 1991, pushed for dogs to be one of the first species with a whole genome sequence, in part because of the potential for using dogs as a model to understand how genetics shapes behavior (30). In 2005, the first dog genome, of a female boxer named Tasha, was published (14) (~2.4 billion nucleotides), making dogs just the fourth mammalian species to have its genome mapped [after human (29), mouse (31), and chimp (32)].

With the completion of that first dog genome, scientists could start using a powerful new approach, genome-wide association study, or GWAS (33), to discover which genetic changes explained differences in appearance, behavior, and disease risk in dogs. GWAS compares individuals with and without a trait at thousands (and now millions) of different positions across the whole genome, looking for the changes that are correlated (or associated) with the trait (34) (**Figure 2**).

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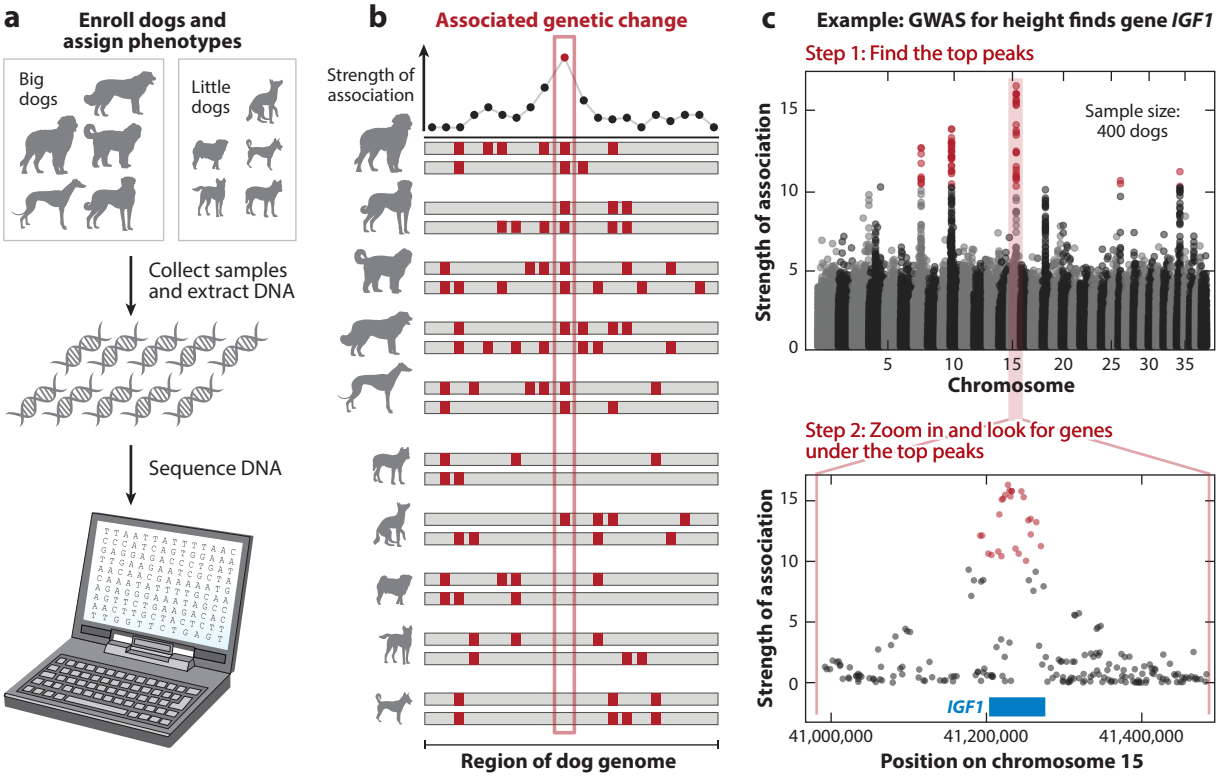

Figure 2

Finding genes for traits using a genome-wide association study (GWAS). (*a*) Steps for making a GWAS data set. (*b*) To find a genetic change associated with a trait, we compare the DNA sequences of all the dogs and look for genetic changes (*red squares*) correlated with the trait of interest. Each dog has two copies of the genome (*gray boxes*)—one inherited from each parent. The genetic change in the middle is more common in big dogs (*outlined in red*) and scores as more associated. (*c*) Example using real data from a GWAS of height with 400 dogs (65). The strongest peak of association is on chromosome 15 and sits over the gene *IGF1*. Dog silhouettes from phylopic.org.

The use of GWAS in humans revealed that many earlier genetic discoveries were suspect. In the past, a common approach for connecting genes with traits involved testing just a small number of genes, which reduced the number of independent tests and allowed scientists to find significant associations in small samples. Most of these associations, however, have failed to replicate in GWAS, including those involving many genes implicated in complex behavioral or neuropsychiatric traits (35–37). Even though this "candidate gene approach" is now widely discredited (38), many of the genetic tests used in dogs today are based on such studies (39).

The first GWAS in humans was published in 2005 and focused on macular degeneration (40), followed just two years later by the first GWAS in dogs, which implicated the *MITF* gene in white coat color (41). Early dog GWAS focused on traits and diseases that are shaped by changes in just one or a few genes (so-called simple or Mendelian traits), including coat color and type (42–45), body size (46–48), and diseases like progressive blindness (49). These early studies confirmed that the limited genetic diversity of dog breeds makes it easier to find regions of the genome associated with traits (50, 51). However, these regions tended to be large, with thousands of genetic differences between dogs, compounding the already huge challenge of pinpointing exactly which genetic changes cause the trait (52).

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Following the success of these early GWAS, scientists started using the approach to explore more complex traits. A complex trait is any trait or disease that is shaped by many genetic changes in conjunction with the environment. Each genetic change has a relatively small effect on the phenotype (the observable trait being measured). The more complex a trait is, the more challenging it is to genetically decipher (53). Genetic studies of complex traits in dogs have produced significant associations, but generally these explain just a small fraction of the heritability, suggesting many more associations have yet to be found.

3.3. Behavioral Genetics in the Genomic Era

More than 60 years ago, Scott & Fuller recognized the potential of dogs as a model system for behavioral genetics. Today, we have the technology to discover the small genetic changes that help shape behavior, personality, and the risk of developing neuropsychiatric diseases. Some studies have tentatively implicated several genes in canine behavioral traits, although the exact genetic changes responsible, and their mechanism of action, are not known. Most studies in dogs are still underpowered for complex traits like behavior, but this is changing quickly.

The most effective way to increase the power of genetic association sizes is to increase the sample sizes enormously (54). For example, the first GWAS of obsessive compulsive disorder (OCD) in humans failed to find any significant associations with a sample size of approximately 7,000 people (55), but a subsequent GWAS with 2,098,077 people found 30 significant associations that give insight into the neurobiology of this hard-to-treat disease (56). GWAS in dogs usually requires somewhat smaller sample sizes than in humans (57), especially when focused on a single breed. A GWAS of a related disease, canine compulsive disorder, found one significant association in a study of 160 Doberman pinscher dogs (58). The more complex a trait is, the larger the sample size required. The gene *MITF* was implicated in white coat color in a GWAS with just 19 dogs (41), but finding genes associated with bone cancer (59), compulsive behavior (58), and clinical measurements (60) required hundreds of dogs.

The recent advent of large-scale dog genomics will transform canine behavioral genetics in the coming years. Genomic resources for dogs continue to advance; the Dog10K project has sequenced 1,987 canids worldwide, including various breeds, village dogs, niche populations, and extended pedigrees, to capture the breadth of genetic diversity in dogs and in reference to other canid species (61, 62). The Mars PetCare Biobank project is sequencing genomes from up to 10,000 pet dogs (63). New initiatives, such as the Dog Aging Project (64) and Darwin's Ark (65), are assembling and sharing data sets of thousands of deeply phenotyped and sequenced dogs. Early work from Darwin's Ark showed that with a cohort of approximately 2,000 dogs, of both mixed- and single-breed ancestry, scientists could identify parts of the genome associated with both specific behaviors (e.g., howling frequency and getting stuck behind objects, which is associated with canine cognitive dysfunction) and broader behavioral factors such as human sociability (65).

However, to realize the full potential of dogs as a natural model for behavioral genetics, even closer collaboration among scientists in the canine research community will be essential. In the human genetics community, it took scientists just a few years to realize that their GWAS sample sizes would need to be far larger than they had anticipated, and that assembling such massive sample sets was beyond the reach of any one group (66). Large consortia are now a defining feature of human genetics. The largest study completed to date has 5.3 million participants and found 7,209 genomic regions that influence human height (67).

A second challenge, however, is figuring out how we collect informative canine behavioral data. In GWAS, we test for genetic changes that correlate with a phenotype. Thus, how that phenotype is defined can have a huge influence on how successful the study is. If the phenotype is determined almost entirely by environment (for example, musical preferences), a GWAS will fail. Conversely,

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if we can define a phenotype that is strongly influenced by genetics, we can make a GWAS more powerful (68). Thus, genetic studies of behavioral traits face a major challenge: How do we define our phenotype such that it is highly heritable and can be measured accurately in many individuals? Addressing the historic divisions between ethology and comparative psychology could help answer this question.

4. MEASURING BEHAVIOR FOR GENETIC STUDIES IN DOGS

4.1. The Challenge of Selecting a Phenotype

Some phenotypes are shaped primarily by an individual's experiences during their lifetime, and less useful for genetic studies, whereas others are strongly influenced by inherited genetics. Whether a phenotype is strongly or weakly attributable to genetics often hinges on how it is measured.

4.1.1. Measuring heritability. Heritability is a statistical measure indicating how much trait variation within a population can be attributed to genetic differences. Importantly, heritability does not describe how much of a trait is due to genetics in an individual. Heritability is also not a fixed property of a trait; it can vary significantly between different populations. A trait might be highly heritable in a population with very little environmental variability but much less heritable once that environment starts to vary. Intrinsic behavioral traits, for example, are almost always highly heritable because their development is influenced by environments that are hard to change, such as the intrauterine environment in mammals or the egg environment in birds. In contrast, accommodative behaviors are influenced by environments that tend to vary more. For example, whether a dog forms social bonds easily with humans and dogs is strongly influenced by its social experiences during the first four months of life. Most pet dogs experience similar levels of human social interaction, but their experiences with unfamiliar dogs are far more varied. In the Darwin's Ark data set with approximately 2,000 dogs, human sociability is far more heritable (67% +/− 13%) than dog sociability (19% +/− 7%) (65).

4.1.2. A good phenotype is neither overly broad nor too narrow. When designing a genetic study, it is crucial to choose a phenotype that is heritable (strongly influenced by genetics) within the population being studied, not too broad and not too narrow. An overly specific (narrow) phenotype will miss associations that would otherwise be detectable. For example, dogs with compulsive disorders can exhibit a range of stereotypical behaviors (e.g., tail chasing, fly snapping, and flank-sucking). If a study defined only those dogs with one type of compulsive behavior as cases, say, tail chasing, it would have less power to detect associations to compulsive behavior risk more broadly by including the entire range of behaviors. Conversely, an overly broad phenotype, even if very heritable, can impede interpretation of the results. In a human GWAS of major depression that used a very broad definition, most of the associations found were not specific to depression but also associated with other psychiatric conditions like neuroticism and schizophrenia (69).

Defining phenotypes in studies of animal behavior often requires considering the context and threshold of the behavior. For example, aggression is a suite of agonistic (distance increasing) behaviors used in a range of contexts, including foraging (protecting food resources), hazard avoidance (defending against predators or competitors), and reproduction (guarding offspring or competing for mates or territory). In dog behavioral studies, however, aggression is often defined very broadly, with little agreement between studies on what constitutes an aggressive response (70–72). If a dog growls at a child for pulling its tail, is that aggression or simply good communication to prevent aggression? A more specific phenotype for a genetic study could use the ethological practice of focusing on how and when the dog deploys more discrete agonistic behaviors, such as

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the context in which they occur, the threshold for triggering them, and the sequence of behaviors preceding and following a given behavior.

4.1.3. Nonbehavioral traits can influence heritability. Some behavioral phenotypes in dogs are highly heritable because they are in part the result of selection on physical traits that are less genetically complex but change how a dog experiences the world. For example, whether a dog avoids getting wet is highly heritable in the pet dog population (48% +/− 10%) (65). This is likely in part because both size and coat type are highly heritable. A dog is more likely to enjoy water if, over their lifetime, they have found that water is a comfortable place to be. Big heavy dogs, and dogs with double coats that protect their skin from getting wet, are more likely to find the cooling effects and the buoyancy of water to be pleasant, but any dog can have positive experiences with water and become water loving, and vice versa. Among 14,000 dog owners surveyed in the Darwin's Ark project, 11% of owners of Labrador retrievers (a famously water-loving breed of large dogs with double coats) reported their dogs did not like getting wet (65).

4.2. Strategies for Collecting Phenotype Data

The success or failure of a genetic study often hinges on how the chosen phenotype is measured. Studies that require large sample sizes may prioritize ease of collection, but in smaller studies phenotype precision can be more important.

4.2.1. Owner questionnaires support large-scale phenotyping. Questionnaires (particularly in a digital format) are a relatively high-throughput strategy for collecting phenotypes for large behavioral genetics studies. Despite being somewhat subjective, and relying on information provided by people without field-specific expertise, self-reported phenotypes have proven to be extremely effective for exploring the genetic causes of complex traits including psychiatric diseases (56) and personality (73). The increase in sample size usually offsets any loss of power resulting from less precise phenotyping but may produce spurious associations (69). An alternative approach might be to survey a professional who has observed the individual (e.g., in dogs, a veterinary behaviorist), although this is difficult for very large samples.

Relying on questionnaires [and especially on internet-based questionnaires (74)] can significantly impact who chooses to participate, with much higher rates of engagement in more affluent and better-educated communities (75). In dogs, using owner-reported phenotypes can raise additional issues. One common concern is whether owner responses are subject to unconscious bias (e.g., preconceptions about breed), but this can be mitigated by including non-breed and mixed-breed dogs in the study (65). In addition, studies relying on owner questionnaires exclude dogs who do not have close relationships with people (e.g., free-living dogs).

4.2.2. Behavioral tests quantitatively measure behavior. Direct behavioral tests can circumvent human opinion and quantitatively measure how a dog behaves in a given situation. The design of the test and the scoring of the behavioral responses must be accurate and reliable (76), and the results of even a carefully designed test may change when it is implemented in a new environment (77). Direct behavioral tests are common in studies where both the environmental and genetic variability can be carefully controlled, such as studies in laboratory animals or, to a lesser extent, working dog breeding programs. For example, as part of their breeding program, Guiding Eyes for the Blind puts all their dogs through the same battery of behavioral tests (78). Because direct behavioral tests are both time and labor intensive, their use has been limited to smaller-scale studies. This is now changing with new software that uses machine learning–based analysis to automate scoring (79). For example, machine learning analysis of wearable sensor data has been

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used to automatically classify the gait patterns of guide dogs (80). These methods may enable use of direct measures of dog behavior in large GWAS.

4.3. Problems with Using Breed as a Proxy for Individual Phenotypes

Given the difficulty of collecting behavioral phenotypes on individual dogs, dog behavioral studies sometimes use a dog's breed to determine its phenotype. Not only does this allow for large numbers of animals to be quickly assigned phenotypes, it can be applied to dogs that were sequenced previously and had their data deposited into public repositories. The dog's breed is first determined using either the accompanying sample information or from the genetic data itself, and then the dog's behavioral phenotype is assigned based on information about other dogs from that breed. The breed phenotypes are defined in various ways, including asking a dog trainer to score the breeds (81), using the breed groupings provided by the American Kennel Club (72), or quantifying breed average phenotypes based on survey data (82). However, all of these approaches rely on the assumption that breed is a reliable predictor of behavior, which runs counter to more than 60 years of research (25, 83–85). Breed may in fact account for as little as 9% of the behavioral variation in an individual dog (65).

4.3.1. Using breed phenotypes in associations studies. Using breed as a proxy for individual phenotypes in genetic association studies (e.g., GWAS) is particularly fraught because breed itself is a genetic trait. In this study design, breed and phenotype will, by definition, be perfectly correlated. Thus, any genetic changes correlated with breed will appear to be associated with the behavioral phenotype, even if the two are not causally connected. This includes the genetic changes underlying the physical characteristics that make dog breeds distinctive. For example, comparing herding breeds to pointing breeds found an association at one of the genes that controls fur length (*RSPO2*) (86) because all of the herding breeds had the same coat type. Another study identified 20 genetic changes associated with behavior using breed phenotypes and proposed that they could be used to predict behavior problems in dogs through genetic testing (87). However, at least 19 of the 20 variants are in or near genes already known to help determine the physical traits that define breeds, including size, snout shape, curly tail, ear shape, fur texture, and coat color (88). This suggests that the use of breed phenotypes explains these associations, undermining the contention that they can predict behavior. Supporting this, GWAS using phenotypes for individual dogs rather than breeds found these variants are associated with physical but not behavioral traits (88). Furthermore, it is unlikely that a small number of changes could accurately predict complex behavioral traits (89). Despite the questions around this approach, dog genetic tests that claim to predict behavior and personality are already being sold to consumers (90).

4.3.2. Using population phenotypes in studies of dog evolution. Population-level phenotypes, including breed phenotypes, can be useful for exploring dog evolution. In these studies, population phenotypes usually are used to develop hypotheses about why some genes might have signals of selection, rather than to find the genes in the first place. A signal of selection is a distinctive pattern of genetic changes that suggests a gene is changing unusually slowly or quickly in response to environmental pressures. For example, when the genomes of gray wolves, who consume primarily large prey animals, were compared to dogs, who evolved to instead eat a starch-rich diet of human food waste, we found signals of selection at many genes related to starch digestion (6). In a large study of genetically diverse dogs, including breed, mixed-ancestry, and non-breed (e.g., village) dogs and wild canids (wolves and dingoes), researchers first identified a set of genes that looked unusual in a group of related sheep dog breeds that had similar scores on some behavioral metrics. They then showed that an unexpectedly large number of these genes were active in the brain (91).

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5. DOGS AS A MODEL FOR HUMAN DISEASES

One reason why behavioral genetics in dogs interests scientists is because of its potential to contribute to research into human neurobehavioral and psychiatric disorders (92). Studies in dogs have already helped identify genes and pathways involved in human diseases, including congenital disorders, neurological diseases such as narcolepsy, amyotrophic lateral sclerosis, and cancer (93–96). Because dogs have naturally occurring diseases and can also be enrolled in veterinary clinical trials, companion dogs help bridge the gap between preclinical studies in laboratory animals and human clinical trials, leading to successful drug development in conditions ranging from ocular disorders (keratoconjunctivitis sicca) to cancer (lymphoma) (97). However, whereas disease comparisons in other organ systems may be more straightforward, using dogs as a model for behavioral disorders requires that we think critically about how comparable the phenotype is between dogs and humans.

5.1. Strengths and Limitations of Dogs as a Complex Social Model

Dogs are of particular interest as model systems for behavioral traits and disorders because they are, in many ways, much more similar to humans than the common laboratory animal models like mice, zebrafish, and fruit flies. For example, whereas early social development takes approximately four years in humans, it lasts only a few weeks in mice (98, 99). Early social development in dogs, although still much shorter than in humans, is approximately eight times longer than in mice (25). Dogs also share much of their environment with their human owners and are exposed to environmental risk factors for behavioral disorders like air pollution (100). In contrast, laboratory animals live in an artificial environment that lacks these environmental toxins but also lacks the enrichment needed for normal brain and behavioral development (101).

However, we are also limited in the ways we can measure behavior and brain function in pet dogs. Unlike diseases of the body, neuropsychiatric diseases in humans often lack specific biological indicators and usually are defined and diagnosed by subjective self-reported symptoms, including thoughts and emotions (102). We cannot measure a dog's thoughts and emotions. For example, canine compulsive disorder is not called canine OCD because dogs cannot tell us if they experience the obsessive component of unwanted or intrusive thoughts, impulses, or images. Invasive experimental approaches used in laboratory mice are not feasible in dogs. For example, optogenetics, which has been used to identify brain pathways involved in compulsive behaviors (103), requires first genetically engineering animals that produce a light-activated molecule in particular parts of their brain and then beaming light into their brain with fiber-optic strands under different experimental conditions (104).

5.2. The Trap of Human Exceptionalism

Despite the strengths of dogs as comparative models, research in this field has been slowed by the narrative of human exceptionalism—the common belief that human brains and cognition, and thus by extension human brain diseases, are fundamentally unique among all species. Other species are undervalued as behavioral models, despite there being little data to support this belief in human exceptionalism. In reality, the human brain is a scaled-up primate brain with the same cellular composition (105), and human cognition is not markedly different from that of other species (106). When people have tested claims of human exceptionalism through behavioral tests of other species, we rarely come out as unique (107). For example, it was thought that humans were the only species that showed mental time travel (thinking about our past and future) (108), but the firing of location-specific place cells in the hippocampus of rats suggests they not only remember recent experiences in mazes but also imagine alternate routes (109). Human cognition is distinctive

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in that we (like all other species) have evolved to fit a specific ecological niche. Studying complex human behaviors in other species requires first assessing what the comparable behavior is in the other species and how it models the human disease.

5.3. Addressing the Translational Gap in Psychiatric Clinical Trials

Drug candidates emerging from laboratory animal experiments have a high rate of failure in human clinical trials. This translation gap has resulted in skepticism about the use of nonhuman animals as models for neuropsychiatric diseases (110, 111). Although there are many reasons that clinical trials fail, sometimes the animal model is not measuring the behavioral phenotype we assume it is measuring. The pathological overgrooming seen in mouse models for OCD may be more similar to trichotillomania, a related hair-pulling disorder (112). A marble-burying rodent model used to screen drugs for OCD has the same response to various drugs that in humans produce different responses (113, 114). When developing a new model, we need to first assess its validity: the extent to which it accurately reflects or assesses what it is intended to measure.

5.3.1. Establishing validity of dog behavioral models. To determine whether given phenotypes in dogs (or any model animal system) are useful for investigating a human disease, we need to first establish the validity of the test. A valid test for a neurobehavioral or neuropsychiatric disorder in dogs can lead to biological insight as a model for human disease. Validity is not inherent to a particular test but rather differs depending on the context (and specific biological question) that the test is being used to investigate. Therefore, a test that is valid in humans is not necessarily valid in dogs.

There are four main types of validity (face, content, predictive, and construct). Face validity assesses whether the test appears to a nonexpert to measure what it is supposed to measure. This type of validity is the simplest and helps to avoid superficial comparisons. For example, the forced swim test in rodents is not a valid model for human depression because it lacks face validity (115). A behavioral phenotype in animals should not be considered analogous to a human condition just because it shares a few behavioral characteristics that in humans are not specific to that disease.

Content validity assesses whether a test is relevant to, and representative of, the phenotype it claims to measure based on experts and literature. For example, the "looking back" behavioral test in dogs (whether a dog looks back to a familiar human when they cannot solve a puzzle) measures how much the dog relies on social cues (116), but this is not sufficient to make it a valid model for the hypersociability phenotype of Williams–Beuren syndrome in humans. People with Williams–Beuren syndrome are hypersociable because they have difficulty reading social cues (117, 118). Thus, the looking back behavioral test has no content validity as a model for Williams–Beuren hypersociability.

Predictive validity refers to whether a test can predict the outcomes of another test. This includes whether a treatment that led to the desired response in a model species also has the expected treatment effect when applied in humans, or vice versa. For example, many of the same selective serotonin reuptake inhibitors (SSRIs) and tricyclic antidepressants are used to treat anxiety disorders in both humans and dogs (92), with similar response rates in the two species. This suggests dogs have predictive validity as a model for therapeutic outcomes from these classes of drugs in treating anxiety. Having predictive validity does not mean the behaviors are the same, just that the outcome of the therapy is similar (i.e., less impairment).

Construct validity measures how well a test reflects an abstract variable, like memory or sociability. It is established by correlating the test with related measures (convergent validity) and unrelated measures (divergent validity). For example, human attention-deficit hyperactivity disorder questionnaires adapted for dogs (119, 120) would show construct validity if dogs rated as

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high-scoring by their owners on this questionnaire were also determined to have high impulsivity and hyperactivity from another method (such as assessments from animal behaviorists).

5.3.2. Using dogs as one model among many. Behavioral genetics in dogs can offer valuable insights into human neuropsychiatric disorders if we can overcome perceptions of human exceptionalism to acknowledge the similarities stemming from our shared ancestry. However, this also requires establishing the validity of the dog model. Given the limitations of working in pet dogs, they are not a replacement for laboratory mice but a powerful addition. All models are imperfect, but integrating genetic studies across multiple species allows them to complement each other.

For example, human OCD is a severe, debilitating disease without effective treatment options. There is a critical need to fill the gap in understanding its genetic and environmental causes (121). Canine compulsive disorder shares key features with human OCD, including spontaneously occurring abnormal repetitive behaviors (face validity) (122), similar responses to treatment with SSRIs (predictive validity) (123), and similar changes in their brains (e.g., altered gray matter volume) (construct validity) (124, 125). Dog models for OCD are now being integrated with laboratory mice model patient cohorts to identify common genetic risk factors for repetitive behaviors (126). Ultimately, finding shared pathways between dogs, humans, and other species is a promising avenue for developing new, and desperately needed, therapeutics (121).

6. THE WAY FORWARD

6.1. Addressing the Historic Dichotomy Between Ethology and Comparative Psychology

Canine behavioral genetics has advanced enormously since its inception in the early twentieth century. We are now poised to start addressing fundamental questions about how genetics shapes behavior. How does a tiny change in DNA create a dog who retrieves (given the right environment)? The addition of biologically based animal behavior to the psychology-based methodologies already in use will help us clarify the questions we ask and the tests we use to answer them. It will also shift the focus of dog research away from behaviors humans find problematic and toward a biological framework linking genetics to behavior. We also need to keep looking beyond breed dogs to the full spectrum of dog behavioral and genetic diversity. The complex and ancient behavioral traits that make dogs unique long predate the breeds, which represent just a tiny portion of dog diversity.

An evolutionary perspective will provide better phenotyping for behavioral genetics. Evolutionary history can provide guidelines for how specific or broad a trait should be to maximize our power to detect associations. Using the ethological technique of measuring things like the frequency, context, and sequence of discrete behaviors in questionnaires can reduce bias and increase the likelihood of finding meaningful heritable phenotypes. Modeling this type of measurement in direct behavioral tests can improve measure reliability and validity, focus attention on heritable phenotypes, and increase the chance of finding associations.

6.2. Overcoming Assumptions of Human Exceptionalism

With behavioral traits and disorders, the value of dogs as a natural model for humans is undermined by (often unsupported) assumptions of human exceptionalism. This limits the number of researchers working in canine behavioral genetics and access to funding. For example, pet dogs could be an unparalleled model for investigating the genetics of aging-related brain diseases (64), including Alzheimer's disease. However, after early studies showed that rodent and primate laboratory animals did not develop many of the defining characteristics of Alzheimer's disease, it was

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deemed to be human specific (127), thus making animal models of limited use. Dogs, however, develop a disease that is remarkably like Alzheimer's, with all the defining characteristics occurring at late stages of life, including cognitive impairment, brain atrophy, plaques, and tangles (128). Despite this, the idea that Alzheimer's disease is unique to humans persists (129), potentially slowing the development of new therapies for this devastating disease.

6.3. Scaling Up Dog Genomics

If we succeed in scaling up dog GWAS data sets, we will have the statistical power to investigate dog behavioral phenotypes in unprecedented detail. The larger the data sets, the more questions we can answer. In each well-powered GWAS for behavioral traits, we will detect dozens or even hundreds of genes associated with each behavioral trait, allowing us to discern the neural pathways shaping the phenotype (130). Once we have well-powered GWAS for many behavioral traits, we can analyze shared heritability to determine which traits share genetic risk factors, suggesting shared underlying causes (131). We can also apply a complementary approach, phenome-wide association studies, that looks at hundreds and even thousands of phenotypes at once to find genetic variants that influence multiple traits (132).

Large genomic data sets also open the door for genomic prediction: using an individual's genome to predict whether they are likely to have (or later develop) a trait or disease. Rather than testing for specific associated genetic changes, genomic prediction considers genetic variation across the whole genome. In populations that lack pedigrees, including humans and most dogs, genomic prediction is done with polygenic risk scores developed using machine learning algorithms. Although potentially powerful, this is difficult to implement. A review of 77 polygenic risk score models developed to predict psychiatric disease risk in humans found that few followed best practices in methodology and reporting (133). It also requires training, testing, and validation data sets from an appropriate population. Polygenic risk scores developed using data from one breed will not work well in other dogs (134).

When breeding dogs, a tool similar to polygenic risk scores known as genomic estimated breeding values (GEBVs) can be used to select for desirable traits in a population. GEBVs represent an estimate of the likelihood that an individual will pass a given phenotype on to their offspring, which would allow dog breeders to select for and against behavioral traits more effectively and quickly. Both polygenic risk scores and GEBVs require tens to hundreds of thousands of dogs to be sequenced and phenotyped. To achieve this, data sharing and collaboration between researchers and large-scale breeding programs are essential (78).

6.4. Adding Environmental Context

Gene-by-environment studies are the next frontier in behavioral genetics, and dogs are a potentially powerful model system because they live with humans. Dog owners are willing to provide detailed information about their dog's environment. On top of that, new technologies like passive samplers (135, 136) and wearables offer the opportunity to collect quantitative environmental and behavioral data (137). In gene-by-environment studies, we investigate how the genetics an individual is born with interacts with the environment they develop in to shape their phenotype (138, 139). This is extremely difficult to decipher because both the genetics and the environment vary between individuals. Two genetically identical individuals can have different behavioral phenotypes because they were raised in different environments (140), but how different their phenotypes are is shaped by their genes (141). In dogs, although we know that early environment affects adult behavior (25, 142–144), the genetic factors that shape this interaction are unknown.

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By partnering with dog owners, we may be able to assemble very large data sets with genetic, behavioral, and environmental data to investigate this.

6.5. Moving from Association to Mechanism

Once we have associated a particular genetic change with a behavioral trait, we still need to test whether it is a causal variant (contributed directly to the trait) and then figure out how it does that. This is a huge challenge. Most trait associations are not in genes but instead alter regulatory elements that influence how that gene is used in different types of cells. In fact, even in human genomics, only a handful of genetic changes have been confidently implicated in complex traits. However, functional investigation is crucial for deciphering the complex interplay between genes and environment, for understanding how behavior evolves in response to environmental change, and for applying genetics to better predict and treat behavioral disorders. Fortunately, a huge suite of new technologies facilitate functional studies. For instance, after associating coat pigment patterns in dogs with genetic changes upstream of the gene *ASIP*, in a promoter region known to regulate *ASIP* expression, scientists sequenced RNA (a molecule made when a gene is being used) in dogs with different coat patterns. They discovered that the different versions of the promoter produced different amounts of RNA, resulting in the different coat patterns observed (145).

To speed the discovery of causal variants, scientists make biological catalogs with information on the products of different genes and the locations of promoters and other types of regulatory elements (146, 147). Unfortunately, the information available for dogs is sparse compared to that for humans and laboratory models like mice. Instead, canine scientists will rely on comparative analyses that allow them to use the information from other species. Although some genomic elements specific to the carnivore lineage will be missed, most genes have similar functions in dogs, humans, and mice. For example, after identifying genomic variants and genes associated with different canine ancestral lineages (e.g., herders and terriers), Dutrow et al. (91) leveraged gene expression data sets across human tissues to conclude that most of these genes were expressed in the brain.

Thanks to advances in gene editing technologies, we can now test the effects of genetic changes directly by inserting those changes into the genome of an organism and observing their effect. These methods can be applied to dogs (148), although this is rare for both logistical and ethical reasons. However, we can also use gene editing to evaluate a genetic change by editing cells in culture (cells grown outside of an organism). For instance, to test regulatory variants associated with canine compulsive disorder, scientists introduced them into cultured human cells through a piece of DNA called a reporter construct and measured whether they could activate gene expression (149). This approach can now be performed at large scale using a technique called massively parallel reporter assays, which can test thousands to millions of genetic variants simultaneously (150, 151). Genome editing with CRISPR/Cas9 is a revolutionary new technology that allows scientists to edit DNA with remarkable precision to test the effect of different genetic variants. For example, CRISPR can precisely turn down or turn up gene expression at specific sites in the genome in neurons grown from patients with neurological diseases (152).

6.6. Implications for Dog Welfare

Dogs are a powerful model for behavioral genetics of value to those studying the fundamental biology of neuropsychiatric diseases in humans, and this has the potential to improve dog health and welfare. For example, understanding how genes and environment interact when dogs are young could help dog raisers provide a better environment for their pups. Understanding the biological origins of pathologies like canine compulsive disorder could lead to more effective therapies. With

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predictive genetic tests for behavioral traits, shelters might be able to identify dogs who are prone to developing severe anxiety and target preventative measures. Realizing the potential of this technology will require close collaboration between genetic scientists and professionals working with dogs to ensure any new tools are used effectively and do not do more harm than good (39).

7. CONCLUSION

Over the last 60 years, canine behavioral genetics has expanded from simply measuring heritability to finding promising associations between genes and behavior. As we look to the future, revisiting the past could help address weaknesses in current studies and make them more powerful. Genetic studies are designed to find the changes in DNA that shape heritable behavioral phenotypes. The phenotypes used in many dog genetic studies, however, are often more a reflection of the dog's environment. For example, dog owners might be asked whether a dog is aggressive, as if this were a single problem behavior in dogs. From an evolutionary standpoint, however, aggression describes a suite of behaviors a dog uses to increase distance between itself and another individual, including growling, snapping, and biting. These behaviors occur differently across all functional behavioral categories (hazard avoidance, reproduction, and foraging). Furthermore, they each have contexts, sequences, frequencies, and thresholds that are shaped by evolution. Carefully defining the heritable phenotype in genetic studies is essential. By defining behavior through an evolutionary lens—the purview of ethology—we can make dog genetic studies more powerful and accelerate the translation of genetic associations into a deeper understanding of the origins of complex behavioral traits and disorders in all animals, including both dogs and their human companions.

DISCLOSURE STATEMENT

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